

Symbolic-JEPA: Domain-Adaptive World Models Outperform LLMs at Physics Reasoning

Research Proposal for LossFunk | Vortaz Labs

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1. THE WHAT — Core Claim

Core Claim

We introduce **Symbolic-JEPA (S-JEPA)** — a domain-adaptive world model that operates directly on structured physical symbols rather than pixels. A ~2M-parameter S-JEPA trained in 26 minutes **outperforms frontier LLMs (up to 120B parameters) by 16x in live Unity physics gameplay**, beats classical ML baselines on scientific CFD aerodynamics prediction, and transfers zero-shot to robotic grasping with 28% fewer trainable parameters. This validates the **World Model Rigidity** hypothesis: LLMs cannot learn causal spatial dynamics from token prediction, regardless of scale or domain fine-tuning.

The Surprise: Behavioral Collapse Under Fine-Tuning

The prevailing assumption is that fine-tuning bridges the gap — a sufficiently fine-tuned LLM should learn any domain. We disprove this. After fine-tuning Llama-3.2-1B via Unsloth QLoRA on the exact same 50K physics trajectories as our world model, the LLM suffered **catastrophic Behavioral Collapse**: across 4 distinct Unity levels, it fired at 0.2 degrees and 100% power on virtually every single shot. It had memorized that maximizing the 'power' token produced impact, but could not spatially interpolate trajectory arcs. The S-JEPA, in contrast, adjusted angles with precision (9.3, 25.0, 5.1, 11.5 degrees) by minimizing latent energy gradients. **Scale is not the bottleneck. Architecture is.**

One-Sentence Version

"We demonstrate that even fine-tuned LLMs undergo Behavioral Collapse when forced to execute continuous physics, while a 2M-parameter Symbolic-JEPA transfers across game physics, aerodynamic CFD, and robotic grasping — establishing that structural topology, not parameter scale, determines competence in physical domains."

Alignment with LossFunk — Research Direction 1: AI That Adapts to a Domain

LossFunk Theme	Our Contribution
Sample efficiency	50K trajectories, 26min training — no internet-scale data required
World models	Built Symbolic-JEPA: extends JEPA's latent-prediction principle with an energy-based critic, multi-start actor, and symbolic (non-pixel) perception — a complete autonomous physics agent
OOD generalization	Zero-shot transfer from NACA-4 to NACA-5 airfoils (0.986 cosine sim); cross-domain transfer from games to grasping
Constrained compute	~2M parameters, runs on a single consumer GPU, 26min end-to-end training
Continual/meta-learning	Frozen Predictor/Critic; only perception encoder swapped per domain

2. THE WHY — Fruitfulness

This is **not** a benchmark improvement paper. We identify a **structural failure mode** of LLMs (Behavioral Collapse under continuous physics), propose an alternative architecture class (symbolic world models with energy-based critics), and demonstrate domain transfer across three heterogeneous physical regimes. Each

result opens new research directions.

New Questions This Result Opens

- **Behavioral Collapse universality:** Does this failure mode manifest in every continuous-action domain, or only where actions have non-monotonic outcome relationships?
- **Topological representation:** Can persistent homology (Betti numbers) replace Cartesian encoders to achieve true scale-invariance across domains with varying mesh resolutions?
- **Hybrid LLM+WM systems:** At what decision boundary should a general LLM hand off to a domain-specific world model? Can this be learned end-to-end?
- **Minimal data frontiers:** We used 50K trajectories; can active exploration reduce this to ~1K while preserving the 16x advantage over fine-tuned LLMs?

Who Would Build on This?

- **Robotics labs** (BAIR, IRIS) — our grasping transfer result (28% fewer params, faster convergence) directly informs sim-to-real manipulation
- **Scientific ML groups** — our PLAID AirFRANS result (0.624 MAE, beating Random Forest and all LLMs) extends to molecular dynamics, climate, materials discovery
- **LossFunk's own agenda** — direct empirical validation of the 'adapting AI' thesis with three cross-domain experiments, not one
- **AI safety researchers** — Behavioral Collapse under fine-tuning is a failure mode that invalidates naive 'just fine-tune it' deployment strategies

Crossroads, Not Corridor

This work connects: **Representation learning** (JEPA, energy models) with **Domain adaptation**; **World models** (LeCun's vision) with **Empirical LLM benchmarking**; **Scientific ML / CFD** (fluid dynamics) with **Embodied AI / robotics**; and **Algebraic topology** (persistent homology, via our T-JEPA roadmap) with **Learned physics**.

3. THE HOW — Experimental Design

Research Scope: Three Heterogeneous Physics Domains

We validate Symbolic-JEPA across three radically different physical regimes to prove the architecture learns **generalized structural topology**, not a hyper-specific rule engine. (1) **Science Birds** (2D rigid-body game physics, 164D symbolic state) — our primary testbed with live Unity execution. (2) **PLAID AirFRANS** (scientific CFD, 5D point-cloud features from 2D aerodynamic meshes) — predicting velocity, pressure, density, and turbulent viscosity fields around airfoils. (3) **2D Robotic Grasping** (32D kinematics + force feedback) — freezing the Predictor and Critic trained on games, swapping only the perception encoder. This cross-domain validation is the ultimate test of whether the architecture captures physics itself rather than game-specific mechanics.

Fairness Protocol: Fine-Tuned LLM Baselines via Unsloth QLoRA

To eliminate the 'specialized model vs zero-shot LLM' critique, we deployed **Unsloth 4-bit QLoRA** to fine-tune Llama-3.2-1B on the exact same 50K physics trajectories used to train S-JEPA, with data formatted as instruction-response JSONL. The fine-tuned LLM received identical domain exposure. When it failed (Behavioral Collapse at 0.2 degrees), the failure was unambiguously architectural — not a deficit of domain data or prompt engineering.

Experiment 1: Live Unity Leaderboard (Real-Time Physics Gameplay)

Rank	System	Parameters	Score	Levels Cleared
1	Symbolic-JEPA (ours)	~2M	2400	3/4
2	Llama-3.2-1B (Unsloth FT)	1B	1325	3/4
3	GPT-OSS-120B (zero-shot)	120B	150	1/4
4	GPT-OSS-20B (zero-shot)	20B	125	0/4
5	Llama-3.1-8B (zero-shot)	8B	0	0/4

S-JEPA achieves **16x the best zero-shot LLM** and **1.8x the fine-tuned Llama-3.2-1B**, despite both the fine-tuned LLM and S-JEPA being trained on identical data. The fine-tuned LLM's partial success is revealing: WebSocket logs show it fired at 0.2 degrees with 100% power on nearly every shot, exploiting collateral physics damage rather than learning trajectory arcs. S-JEPA actively scanned latent geometric surfaces, adjusting angles precisely (9.3, 25.0, 5.1, 11.5) to minimize continuous energy gradients (Energy: -2.07, -1.88).

Experiment 2: Scientific CFD — PLAID AirFRANS Aerodynamics

Rank	Architecture Type	Model	MAE	Cosine Sim
1	Latent Variable Model	Symbolic-JEPA (ours)	0.624	0.5214
2	Classical ML	Ridge Regression	0.634	0.5006
3	Classical ML	Random Forest	0.635	0.4833
4	Classical ML	Deep MLP (PyTorch)	0.684	0.4204
5	Language Model	Llama-3.1-8B	693.4	0.0274
6	Language Model	GPT-OSS-120B	885.6	0.0222

S-JEPA achieves **zero-shot NACA-4 to NACA-5 airfoil transfer** at 0.986 cosine similarity, with Energy Critic correlation to OpenFOAM of Spearman rho = 0.509 (p = 8.9e-28). Language models fail catastrophically (cosine sim ~0.02), unable to predict continuous pressure fields from symbolic point clouds.

Experiment 3: Cross-Domain Transfer — Robotic Grasping

Configuration	Trainable Params	Epoch 2 Cos Sim	Final Cos Sim
From scratch (full retrain)	942,592	0.965	0.997
Transfer (frozen Pred/Critic)	676,864 (28% less)	0.966	0.995

Freezing the Predictor and Critic trained on Science Birds and swapping only the 32D grasping encoder, we reduce trainable parameters by 28% while matching full-retrain performance — convergence at Epoch 2 is actually faster. This proves the core engine learned **generalized structural topology**, not game-specific rules.

Addressing Alternative Explanations

The results are not cherry-picked — S-JEPA outperforms across three disjoint domains (games, CFD, grasping) on held-out evaluations. It is not relying on heuristics: the critic achieves Spearman rho = 0.71 against game outcomes and 0.509 against OpenFOAM CFD, and the actor performs real gradient optimization through frozen weights. The LLM failures are not due to poor prompting — fine-tuned Llama-3.2-1B with identical 50K training trajectories still Behavior-Collapsed. The comparison is fair: identical perception pipelines, identical action spaces, identical execution interfaces. S-JEPA is not overfit: it transfers zero-shot across airfoil families and across physical domains.

Architecture: Symbolic-JEPA with Energy-Based Critic

S-JEPA extends JEPA's latent-prediction principle by discarding pixels entirely. Where V-JEPA (visual) uses CNNs/ViTs on image patches, S-JEPA operates on **structured symbols** — 164D object-center vectors (games), 5D point-cloud features (CFD), 32D kinematic vectors (grasping). The **Encoder** maps raw symbols directly to a 256D latent manifold via an MLP stabilized by an EMA target encoder (momentum 0.98) and SIGReg regularization. The **Predictor** (259D to 256D) evolves latent states under action conditioning, encoding domain physics into weights.

The **Energy Critic** enforces physical boundary conditions: states violating continuity (e.g., overlapping masses, invalid pressure gradients) receive high energy. Trained via contrastive margin loss with 2.3x positive-sample oversampling, it achieves Spearman rho = 0.71 against game outcomes. The **Actor** performs multi-start Adam optimization (8 starts x 100 steps) through frozen Predictor and Critic — genuinely searching action space per state, not memorizing. An auxiliary **Decoder** enables interpretability. Total: ~2M parameters.

4. THE SO WHAT — Impact and Depth

Depth Level: Level 3 — General Principle

Level	Description	Our Status
2	Explains WHY world models succeed and LLMs fail	LLMs learn token co-occurrence; S-JEPA learns latent dynamics. Fine-tuned LLMs undergo Behavioral Collapse.
3 (achieved)	General principle across domains	Validated across games, CFD, and grasping. Structural topology transfers; linguistic patterns do not. Scale does not substitute for structure.

Impact Type

- **Conceptual:** Introduces Behavioral Collapse as a named failure mode; disproves 'fine-tuning fixes everything'
- **Practical:** Deployable 2M-parameter architecture for any physics domain in 26min on consumer hardware
- **Methodological:** Fair comparison framework (identical perception + fine-tuned LLM baselines via Unsloth QLoRA)

The Science Version

Title: "Scale Does Not Substitute for Structure: Symbolic World Models Outperform Fine-Tuned LLMs at Continuous Physics Across Game, Fluid, and Robotic Domains"

Headline claim: Language models, even when fine-tuned on identical domain data, undergo Behavioral Collapse when executing continuous physics — collapsing into static policies that exploit single tokens rather than spatial interpolation. A 2M-parameter Symbolic-JEPA trained in 26 minutes outperforms 120B-parameter LLMs by 16x in live Unity physics, beats all classical ML on CFD aerodynamics prediction, and transfers zero-shot to robotic grasping with 28% fewer trainable parameters — establishing that structural topology, not parameter count, determines competence in physical domains.

Future Direction: Topological-JEPA (T-JEPA)

S-JEPA is bottlenecked by fixed-size input arrays (164D for games, 27D for aerodynamics). We propose **Topological-JEPA**: replacing Cartesian encoders with **persistent homology** from algebraic topology. By extracting Betti numbers (connected components, loops, voids) from physical state, T-JEPA achieves absolute scale invariance — whether a fluid boundary has 50 nodes or 50 million, its topology is mathematically identical. This transitions world models from learning coordinate symbols to mastering the invariant shape of physical reality itself.